

Empty Vehicle Repositioning for Autonomous Mobility-on-Demand systems via Coverage Control

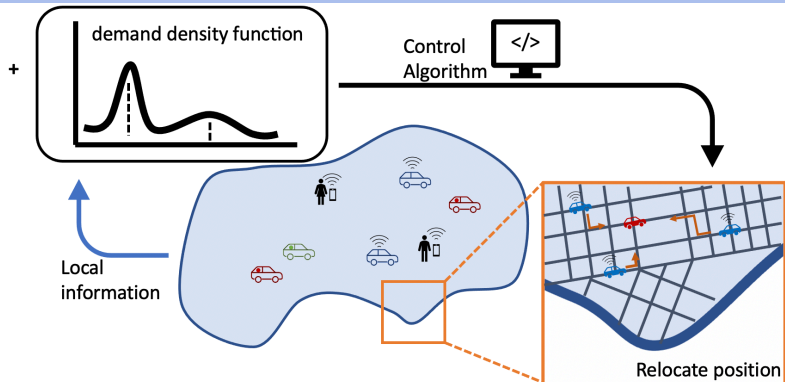
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1 Introduction

Imbalance in the spatial distribution of vehicles:

- Asymmetric origin and destination distributions of trips
- Non-uniform demand for rides in different districts



Proposition: The local maximum of H can be obtained when all x_i are located at centroids (centers of mass, C_{W_i}) of their respective Voronoi cells (W_i), i.e., *Centroidal Voronoi Configuration (CVC)*.

Control Law Formulation:

$$\dot{x}_i = u_i,$$

$$\text{Let } u_i = -k_p(x_i - C_{W_i}), \quad k_p > 0,$$

which steers the agent team to converge to CVC and guarantees the monotonic increasing of H .

3 Methodology II: Idle fleet sizing problem

Motivation:

When Supply > Demand, empty kilometers traveled caused by repositioning result in energy consumption.

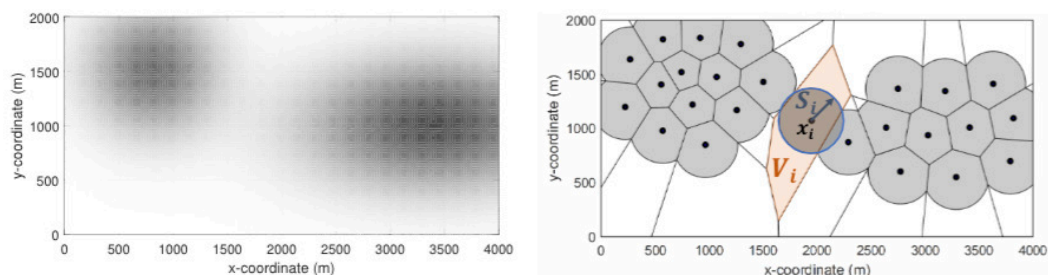
A possible solution:

Select a subset of idle vehicles not to be relocated.

2 Methodology I: Coverage Control for Vehicle Repositioning (CVR)

Voronoi partition:

$$V_i(x_i) = \{q \in \Omega : \|x_i - q\| \leq \|x_j - q\|, j \neq i\}$$



(a) Demand density function

(b) Final configuration

x_i : position of idle AVs,
 r : coverage radius,

S_i : covered area,
 V_i : Voronoi cell,
 $W_i = S_i \cap V_i$.

Two questions:

1) How many AVs should be relocated for different demand levels?

▪ **CVR- α** : $\lfloor n_{idle} \times \alpha \rfloor$ are selected as 'active idle AVs' where $\alpha = 0\%, 20\%, 40\%, 60\%, 80\%, 100\%$.

▪ **CVR-PI**: a proportional-integral controller.

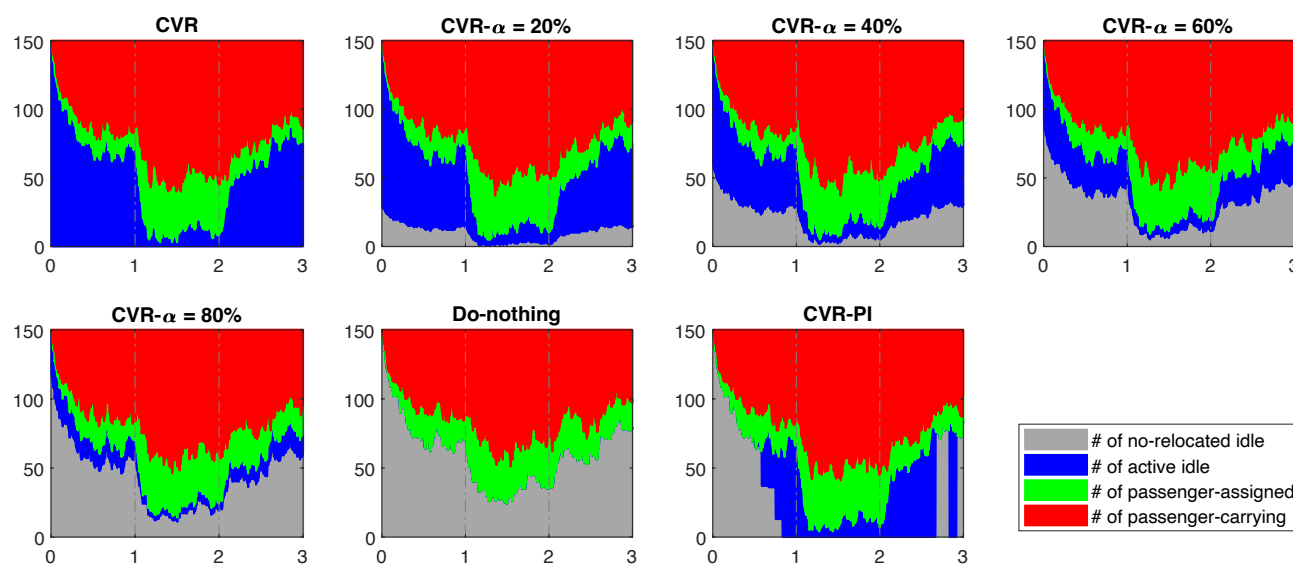
The fleet size is determined on n_{idle} and the average waiting time in last 5 min.

2) Which ones should be chosen for not be relocated?

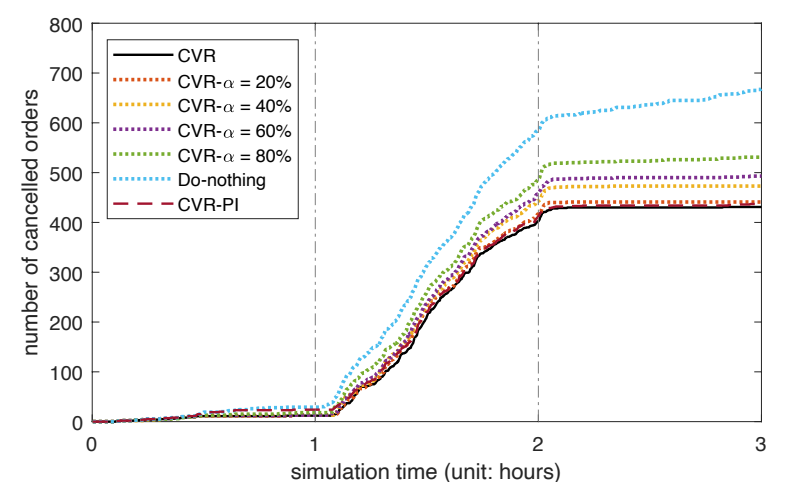
Considering the coverage objective, AVs with smallest value of $J(W_i, x_i)/J(V_i, x_i)$ will be selected as 'no-relocated' as they are around the high demand area, where

$$J(W_i, x_i) = \int_{q \in W_i} \|x_i - q\|^2 \phi(q) dq.$$

$$J(V_i, x_i) = \int_{q \in V_i} \|x_i - q\|^2 \phi(q) dq.$$



(a) No. of vehicles in different states



(b) No. of accumulated canceled orders

4 Conclusion

Our proposed methods dynamically rebalance spatial distribution of vehicles to **serve more trips with less waiting time**. By selecting a subset of idle AVs not be relocated, we can serve almost the same number of requests while **reducing 51.8% empty travelling kilometers** with CVR-PI.

References

The archived submission did not contain a separately labelled references section.